TECHNOLOGY

Comptia Security + 701
Course-end Project

Infrastructure Security

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Objectives

Enhance the security and manageability of its IT infrastructure by implementing a centralized Active Directory (AD) domain using Windows Server 2022. The goal is to streamline user management, enforce robust security policies, and ensure compliance with industry standards such as NIST and CIS, thereby mitigating potential security risks and operational inefficiencies

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Prerequisites

- Active Directory Users and Computers (ADUC)
- Group Policy Management Console (GPMC)
- DNS and DHCP Management Tools
- PowerShell

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Problem Statement and Motivation

Problem Statement:

Enhance the security and manageability of its IT infrastructure by implementing a centralized Active Directory (AD) domain using Windows Server 2022. The goal is to streamline user management, enforce robust security policies, and ensure compliance with industry standards such as NIST and CIS, thereby mitigating potential security risks and operational inefficiencies

Real-World Scenario:
An enterprise struggling with decentralized IT management and inconsistent security policies decides to implement an Active Directory (AD) domain to centralize user administration and enhance security across its network.

Following the AD deployment, the company experienced improved efficiency in managing user accounts, enforcing security measures, and ensuring compliance with industry standards, leading to better data protection and streamlined IT operations.

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Industry Relevance

Skills used in the project and their usage in the industry are given below:

1. **Windows Server 2022 and Windows 10:** For setting up the AD domain and joining client machines to the domain
2. **Active Directory and Group Policy Tools:** Includes ADUC and GPMC for managing OUs, user accounts, and enforcing security policies
3. **Monitoring and Validation Tools:** Event Viewer, RSoP, and PowerShell for auditing, monitoring security logs, and ensuring policy compliance
4. **Network and Compliance Tools:** For configuring network settings and verifying alignment with NIST and CIS standards

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Task (Activities)

1. Setup Active Directory: Deploy Windows Server 2022 as the domain controller and create an AD domain to manage network resources.

2. Client Configuration: Join Windows 10 client machines to the newly created AD domain for centralized management.

3. Organizational Units and Groups: Define 'Executives' and 'Research' OUs, and create role-specific groups within each OU.

4. User Management: Create user accounts, assign them to respective groups, and test login connectivity on Windows 10 clients.

5. Security Policies: Implement NIST and CIS-aligned password policies, configure login hours, enable audit logging, and set account lockout policies.

6. Compliance and Reporting: Generate RSoP reports to demonstrate policy implementation and ensure compliance with security standards.

7. Testing and Validation: Validate user access, review security logs, and audit compliance to ensure all settings are applied correctly..

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Submission Process

You will have to submit the project in one week.

It is recommended to work in the integrated labs, as they have all the required tools available.

The project can be submitted from the assessment tab by clicking the **Submit** button.

Provide the documents mentioned below:

- Source Code in zip
- Database scripts to replicate your database settings
- Screenshots of the outputs

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Thank You

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